

HWA CHONG INSTITUTION
2011 C2 PRELIMINARY EXAMINATION
H1 CHEMISTRY 8872
Answer scheme

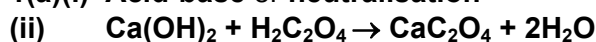
Paper 1 MCQ

1	2	3	4	5	6	7	8	9	10
B	D	D	B	D	D	D	B	A	C
11	12	13	14	15	16	17	18	19	20
A	C	B	A	A	C	D	B	C	B
21	22	23	24	25	26	27	28	29	30
A	D	C	B	B	B	C	A	B	C

Paper 2

Section A

1(a)(i) Acid-base or neutralisation



(iii) ionic bonding (between Ca^{2+} and $\text{C}_2\text{O}_4^{2-}$)
 covalent bonding in $\text{C}_2\text{O}_4^{2-}$ (between C and C & between C and O)



(ii) No of mol of Z (CaCO_3) = no of mol of CO
 $= \frac{0.028}{28.0} = 0.0010$

Mass of Z (CaCO_3) obtained = $0.0010 \times 100.0 = 0.10 \text{ g}$

(c)(i) $\frac{\text{no of mol of } \text{K}_2\text{C}_2\text{O}_4}{\text{no of mol of } \text{KMnO}_4} = \frac{\frac{25.0}{1000} \times 0.050}{\frac{22.70}{1000} \times 0.022} = 2.5 \text{ or } \frac{5}{2}$

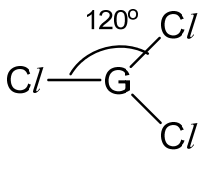
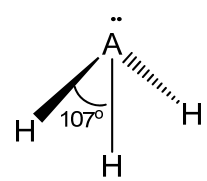
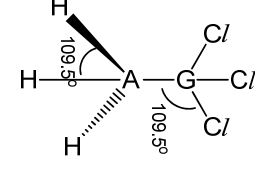


2(a) J has an extra quantum shell; the valence e's are further away from nucleus and less strongly attracted and thus a lower IE

(b)(i) IV

(ii) HO_2 or SiO_2

(c)

		
chloride of G	hydride of A	product formed

(d)

	Chloride of G	Chloride of J
pH of solution formed when added to water	3	1 or 2
Equation to show reaction with water (if any)	$\text{GCl}_3 + 6\text{H}_2\text{O} \rightarrow [\text{G}(\text{H}_2\text{O})_6]^{3+} + 3\text{Cl}^-$ or $\text{GCl}_3 + 3\text{H}_2\text{O} \rightarrow \text{G}(\text{OH})_3 + 3\text{HCl}$	$\text{JCl}_3 + 3\text{H}_2\text{O} \rightarrow \text{H}_3\text{JO}_3 + 3\text{HCl}$ $\text{JCl}_5 + 4\text{H}_2\text{O} \rightarrow \text{H}_3\text{JO}_4 + 5\text{HCl}$

3(a) VII – the biggest increase in IE from 7th to 8th $\Rightarrow$ 7 e⁻s in valence shell

(b)(i) K in Gp VI, M in Gp VIII and N in Gp I of next Period

K: 2– L: 1– N: 1+

(ii) Ionic radius: $K^{2-} > L^{-} > N^{+}$

all 3 ions are isoelectronic while the nuclear charges increases from K^{2-} to L^{-} to N^{+} ; thus weaker forces of attraction between valence e⁻s and nuclear charges

(c)

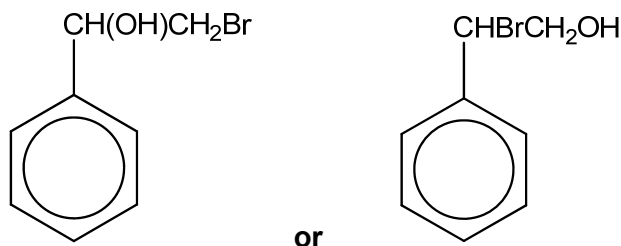
(i) K and L	(ii) K and N.
	$2[N]^{+} \quad [K]^{2-}$

(d)(i) Compound in c(ii): giant ionic lattice with strong electrostatic forces of attraction between N^{+} and K^{2-} ions

Element L – simple molecular (L_2) with weak intermolecular van der Waals (dispersion) forces

(ii) No, compound in c(ii) is ionic; ionic compounds do not conduct electricity in solid state since the charges are immobile; however they conduct electricity when molten or dissolved in water

4(a)(i)



(ii)

Step	reagents and conditions
I	excess concentrated H_2SO_4; 170 °C
II	acidified $\text{KMnO}_4(\text{aq})$; heat
III	acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$; distil
IV	Cl_2, FeCl_3 (or AlCl_3); heat

(b)

	1-phenylethanol	2-phenylethanol
Reagents and conditions	aqueous alkaline $\text{I}_2(\text{aq})$; warm	
Observations	yellow precipitate	no yellow precipitate

SECTION B

5(a)(i)

Mr of propan-2-ol = 60; let mass of propan-2-ol needed = m g

$$\text{heat from burning of propan-2-ol} = 0.80 \times \frac{m}{60} \times 2006 \times 10^3 \text{ J}$$

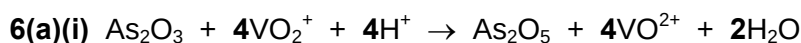
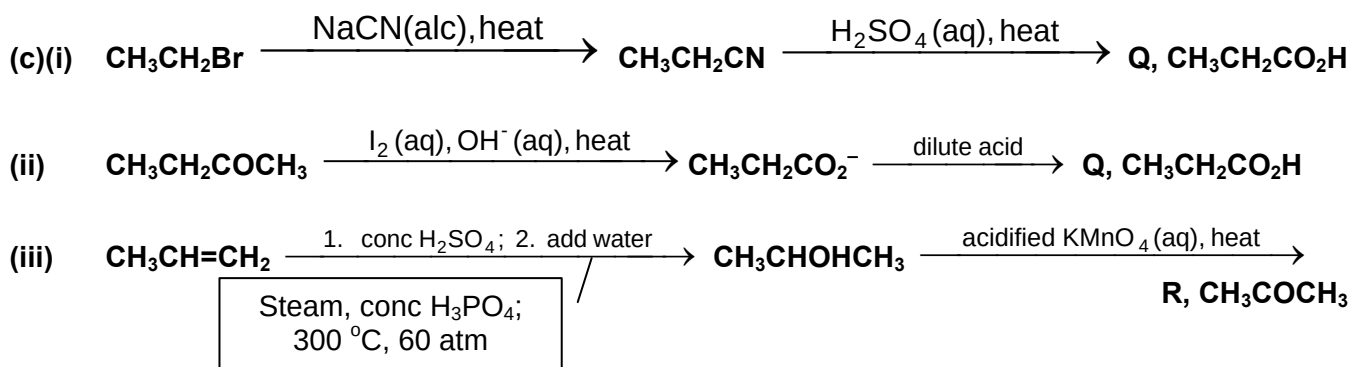
$$\text{heat gained by water} = 1000 \times 4.2 \times 70 \text{ J}$$

$$\text{heat from burning of propan-2-ol} = \text{heat gained by water}$$

$$0.80 \times \frac{m}{60} \times 2006 \times 10^3 = 1000 \times 4.2 \times 70$$

$$m = 11.0 \text{ g}$$

- (ii) I **heat evolved** when 1 mole of propan-2-ol is **completely burnt** in excess oxygen
- II enthalpy change of formation of 1 mol propan-2-ol from its elements in their standard states
- III $\Delta H_c = \Delta H_c(\text{products}) - \Delta H_c(\text{reactants})$
 $= 3(-393) + 4(-285) - (-2006)$
 $= -313 \text{ kJ mol}^{-1}$
- (b) - P, $\text{C}_6\text{H}_{12}\text{O}_2$, likely an ester (is neutral)
 - P undergoes hydrolysis when heated with dilute acid
 - P undergoes oxidation when heated with $\text{MnO}_4^- (\text{aq})/\text{H}^+$
 - Q, $\text{C}_3\text{H}_6\text{O}_2$, give CO_2 (effervescence) with $\text{Na}_2\text{CO}_3 (\text{aq}) \Rightarrow$ is an acid (has $-\text{CO}_2\text{H}$ group)
 - Q is $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$
 - R, $\text{C}_3\text{H}_6\text{O}$, undergoes condensation with 2,4-DNPH & NOT Fehling's solution $\Rightarrow$ is a ketone
 - R is CH_3COCH_3
 - P is $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}(\text{CH}_3)_2$ (as propan-2-ol is oxidised to propanone after hydrolysis by acid)



- (ii) Half-life of a reaction is the time taken for the concentration of a reactant to decrease to half its initial value.

Initial rate of reaction is the rate at the start of a reaction, i.e., when time = 0.

- (iii) - Half-life for the experiment 2 is almost constant of 48 min
 $\therefore$ reaction is first order with respect to VO_2^+

initial rates:

Experiment No	$[\text{As}_2\text{O}_3] / \text{mol dm}^{-3}$	initial rate / $\text{mol dm}^{-3} \text{ min}^{-1}$
1	0.05	$0.0030 / 85 = 3.52 \times 10^{-5}$
2	0.10	$0.0040 / 58 = 7.02 \times 10^{-5}$

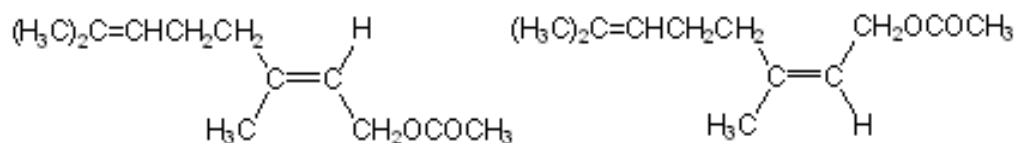
- Initial rate approximately doubled when $[\text{As}_2\text{O}_3]$ doubled
 $\therefore$ rate $\propto [\text{As}_2\text{O}_3]$, i.e., reaction is first order with respect to As_2O_3

$$\text{Rate} = k [\text{As}_2\text{O}_3] [\text{VO}_2^+]$$

- (iv) initial rate for experiment 2 = $0.0030 / 85 = \underline{3.52 \times 10^{-5}} \text{ mol dm}^{-3} \text{ min}^{-1}$

$$k = \frac{\text{rate}}{[\text{As}_2\text{O}_3][\text{VO}_2^+]} = \frac{7.02 \times 10^{-5}}{0.10 \times 0.0050} = \underline{0.140 \text{ mol}^{-1} \text{ dm}^3 \text{ min}^{-1}}$$

(b)(i) geometrical isomerism



(ii) - head-on overlap of 2 orbitals to form a σ bond:

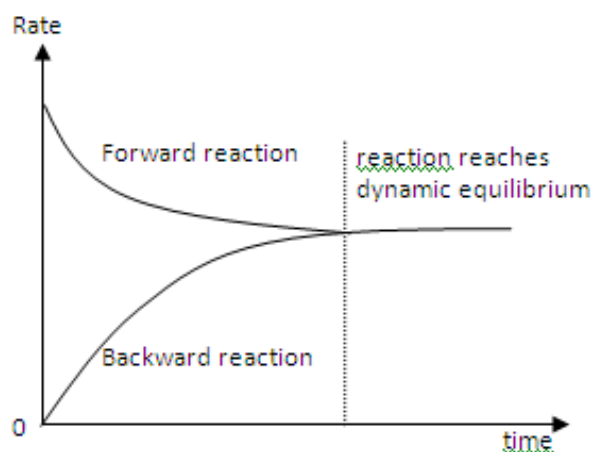


- side-on overlap of 2 orbitals to form a π bond:

(iii) I **Addition**
 $(\text{CH}_3)_2\text{CHCHBrCH}_2\text{CH}_2\text{CH}(\text{CH}_3)\text{CHBrCH}_2\text{OCOCH}_3$
 or $(\text{CH}_3)_2\text{CBrCH}_2\text{CH}_2\text{CH}_2\text{CBr}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{OCOCH}_3$

II Hydrolysis
 $(\text{CH}_3)_2\text{C}=\text{CHCH}_2\text{CH}_2\text{CCH}_3=\text{CHCH}_2\text{OH}$
 CH_3CO_2^- or $\text{CH}_3\text{CO}_2^- \text{Na}^+$

7(a)(i)



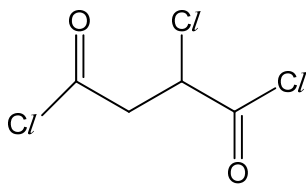
$$(ii) \quad I \quad K_c = \frac{[SO_3]^2}{[SO_2]^2 [O_2]}$$

II	2SO₂(g)	+	O₂(g)	⇌	2SO₃(g)
initial amt / mol	2		1		0
change	-1.60		-0.80		+1.60
eqm amt / mol	0.40		0.20		1.60

$$[\text{SO}_2] = 0.20 \text{ mol dm}^{-3}; [\text{O}_2] = 0.10 \text{ mol dm}^{-3}; [\text{SO}_3] = 0.80 \text{ mol dm}^{-3}$$

$$K_c = \frac{0.80^2}{0.20^2 \times 0.10} = 160 \text{ mol}^{-1} \text{ dm}^3$$

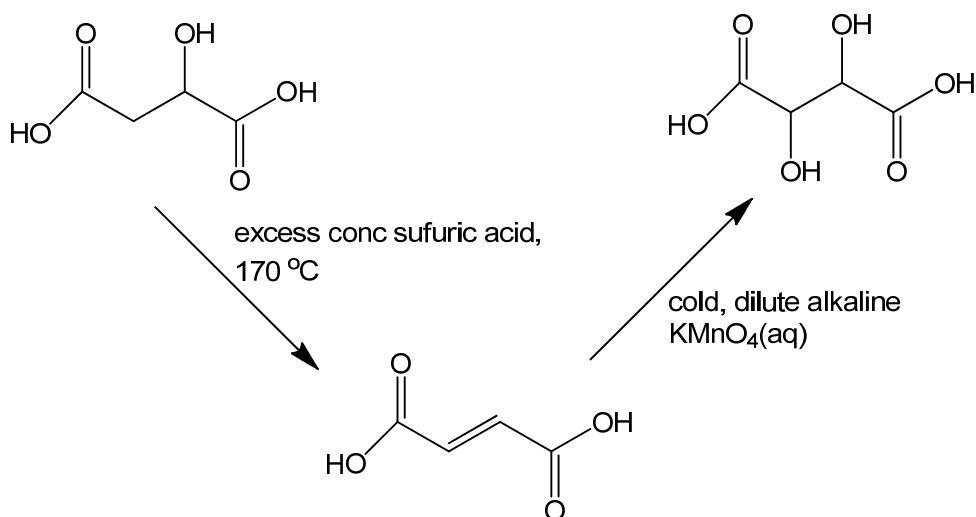
(b)(i)



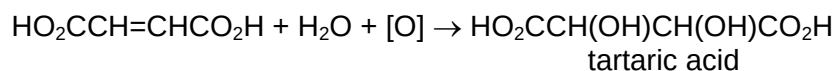
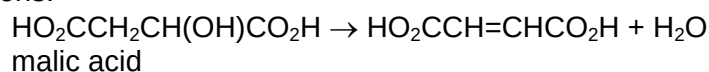
(ii)

Reagents and conditions	Observations	
	Citric acid	Malic acid, tartaric acid
Heat with acidified $\text{KMnO}_4(\text{aq})$	No colour change	purple to colourless
Heat acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$	No colour change	orange to green

(iii)

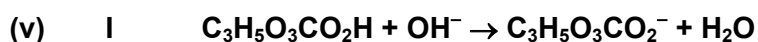


Equations:

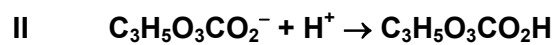
(c)(i) $[\text{H}^+] = 0.00316$ (or 3.16×10^{-3}) mol dm^{-3}

$$\text{(ii)} \quad \frac{\text{no of mol of malic acid}}{\text{no of mol of NaOH}} = \frac{\frac{20.0}{1000} \times [\text{malic acid}]}{\frac{23.30}{1000} \times 0.15} = 1$$

$$[\text{malic acid}] = 0.175 \text{ mol dm}^{-3}$$

(iii) Malic acid is a weak acid and only dissociates partially in water; thus $[\text{H}^+] \ll [\text{malic acid}]$ (iv) **Phenolphthalein**; pH at the end-point of weak acid-strong base is **above 7** which corresponds to **working range** of phenolphthalein

As **original** $[\text{C}_3\text{H}_5\text{O}_3\text{CO}_2\text{H}]$ & $[\text{C}_3\text{H}_5\text{O}_3\text{CO}_2^-]$ are **large**, **change in pH** of solution is **insignificant**.



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